

HDOC DAY 43

PROBLEM

Write a menu-driven C program to perform the following operations using separate functions:

- Find the longest word in a sentence.
- Check if one string is a rotation of another.

CODE

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

#define MAX 200

// Function to find the longest word in a sentence
void longestWord(char str[])
{
    char word[MAX], longest[MAX];
    int i = 0, j = 0, maxLen = 0;

    while (1)
    {
        if (str[i] != ' ' && str[i] != '\0' &&
            str[i] != '\n' && str[i] != '\t')
        {
            word[j++] = str[i];
        }
    }
}
```

```

    }
    else
    {
        word[j] = '\0';

        if (j > maxLen)
        {
            maxLen = j;
            strcpy(longest, word);
        }

        j = 0;
    }

    if (str[i] == '\0' || str[i] == '\n')
        break;

    i++;
}

if (maxLen > 0)
    printf("Longest word: %s\n", longest);
else
    printf("No word found.\n");
}

```

// Function to check whether two strings are rotations

```
int isRotation(char str1[], char str2[])
```

```

{
    char temp[2 * MAX];
    int len1 = strlen(str1);
    int len2 = strlen(str2);

    // Remove newline characters
    if (len1 > 0 && str1[len1 - 1] == '\n')
        str1[--len1] = '\0';

    if (len2 > 0 && str2[len2 - 1] == '\n')
        str2[--len2] = '\0';

    if (len1 != len2)
        return 0;

    strcpy(temp, str1);
    strcat(temp, str1);

    return strstr(temp, str2) != NULL;
}

```

```

int main()
{
    int choice;
    char sentence[MAX];
    char str1[MAX], str2[MAX];

    do

```

```

{
    printf("\n----- MENU ----- \n");
    printf("1. Find the longest word\n");
    printf("2. Check string rotation\n");
    printf("3. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    getchar();

    switch (choice)
    {
        case 1:
            printf("Enter a sentence: ");
            fgets(sentence, MAX, stdin);
            longestWord(sentence);
            break;

        case 2:
            printf("Enter the first string: ");
            fgets(str1, MAX, stdin);

            printf("Enter the second string: ");
            fgets(str2, MAX, stdin);

            if (isRotation(str1, str2))
                printf("The strings are rotations.\n");
            else
                printf("The strings are not rotations.\n");
    }
}

```

```
break;
```

```
case 3:
```

```
printf("Exiting program.\n");
```

```
break;
```

```
default:
```

```
printf("Invalid choice! Try again.\n");
```

```
}
```

```
} while (choice != 3);
```

```
return 0;
```

```
}
```